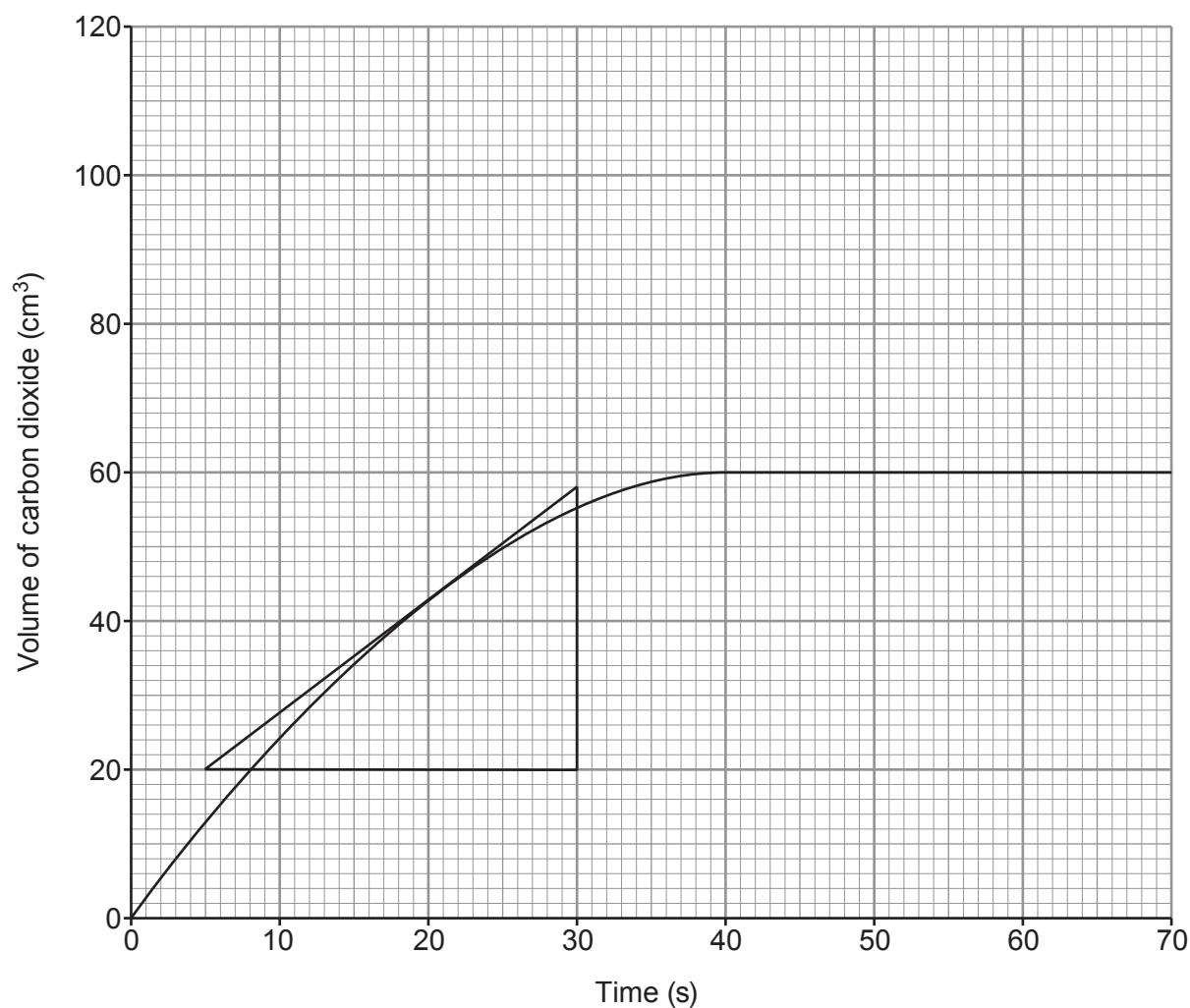


11. The graph shows the volume of gas produced in a reaction between 0.40 g of calcium carbonate powder and excess 1 mol/dm<sup>3</sup> hydrochloric acid.



- (a) Use the tangent drawn to calculate the rate of the reaction at 20 s.

[2]

Rate = ..... cm<sup>3</sup>/s

- (b) (i) On the same grid, sketch the graph that would be seen if the reaction were repeated with 0.60 g of calcium carbonate powder and acid of the same concentration and still in excess. [2]

- (ii) Explain the graph that you have sketched using your knowledge of particle theory. [2]

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**END OF PAPER**

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